Arslan Khalid

Machine Learning Engineer

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Education

B.Sc. Electrical Engineering
University of Gujrat
CGPA: 3.12
Nov 2020 – Jul 2024

Relevant Courses:
- Advanced ML Algorithms by Stanford University
- Supervised Machine Learning by Stanford University
- Django Web Framework by Meta

Skills

Languages:	Python · C++ · JavaScript
Libraries/Frameworks:	TensorFlow · PyTorch · Hugging Face · LangChain · OpenCV · Streamlit
Machine Learning:	NLP · Computer Vision · Reinforcement Learning (RL) · RAG Models
Version Control:	Git · GitHub · GitLab
Development:	Clean Code · Deploying ML Models · Building Streamlit Apps
Soft Skills:	Team Player · Problem Solving · Critical Thinking · Communication Skills

Experience

- **GulzarSoft (On-site)** — Gujrat, Pakistan
  Machine Learning Engineer — *Oct 2024 – Present*
  - **Car Recommendation System**: Designed and implemented a collaborative filtering model for a car recommendation system, enhancing recommendation accuracy by 30%.
- **Coders Launchpad (Remote)** — Dubai, UAE
  Machine Learning Intern — *Nov 2023 – Feb 2024*
  - **Voice-to-Voice Conversion Model**: Assisted in training a voice-to-voice conversion model for the Quran Project using RVC, contributing to high-quality audio synthesis.
  - **Audio Synthesis**: Enhanced audio quality for Quran recitations, improving accuracy and naturalness in voice conversion.
- **Intelligent Systems Lab (On-site)** — Gujrat, Pakistan
  Machine Learning Intern — *Sep 2023 – Oct 2023*
  - **Facial Recognition Model**: Developed a facial recognition model using TensorFlow to match ID card images with live photos, improving identification accuracy.
  - **Image Matching**: Leveraged machine learning techniques to accurately match facial features between static and live images, streamlining identification.

Projects

- **Stereo Vision Navigation with RL**: Developed an autonomous quadruped robot for path planning and navigation using reinforcement learning. Integrated obstacle detection and avoidance with advanced machine learning algorithms.
- **Jarvis — Real-Time AI Voice Assistant**
  LiveKit, Python, WebSockets, Wake-Word Detection
  - Architected a persistent, low-latency AI voice assistant leveraging LiveKit real-time infrastructure.
  - Trained and deployed custom wake-word detection models, optimizing streaming connection latency to achieve near-instantaneous conversational responsiveness.
  - Bridged continuous audio streaming with backend AI processing pipelines, enabling seamless and natural user interactions.

- **Cortex — LLM Reasoning Visualization Tool**
  React, TypeScript, Python, FastAPI, WebSockets
  - Designed and deployed a full-stack software system dedicated to tracking, logging, and visually mapping the internal reasoning chains of Large Language Models (LLMs).
  - Built a highly responsive frontend dashboard and robust backend architecture to process and display complex AI inference steps in real-time.
  - Enabled developers to effectively debug, audit, and optimize LLM prompts and agent behaviors through comprehensive visual analytics.
- **AI-FSL — EdTech Curriculum Generator**
  Python, OpenAI API, Structured Outputs, JSON Schema
  - Created an automated lesson plan generator and comprehensive French immersion platform adopted by Canadian school boards.
  - Engineered a specialized generation engine utilizing strict structured parameters and rule-based logic to perfectly align AI outputs with official educational standards.
  - Streamlined the creation of highly specialized educational materials, significantly reducing curriculum preparation time for educators.